

Usability Considerations	Passwords	Look-Up Secrets	Out-of-Band Devices	Single-Factor OTP	Multi-Factor OTP	Single-Factor Cryptographic	Multi-Factor Cryptographic
Typical Usage							
Authenticator availability - authenticators readily in user’s possession	•	•	•	•	•	•	•
Plain language for user-facing text (e.g., instructions, prompts, notifications, error messages)	•	•	•	•	•	•	•
Legibility of user-facing text or text entered by users	•	•	•	•	•	•	•
Unmasked text entry		•	•	•	•		
Support text entry - length of 64 characters, copy and paste	•						
Delayed masking during text entry	•						
Adequate time allowed for text entry	•	•	•	•	•		
Entry errors - Need clear and meaningful feedback	•	•	•	•	•		
Minimum of 10 attempts allowed	•	•	•	•	•		
Remaining allowed attempts - need clear and meaningful feedback	•	•	•	•	•		
Form factor constraints	•	•	•	•	•	•	•
Location and availability of a direct computer interface such as a USB port						•	•
Physical input required (such as pressing a button)						•	
Cryptographic keys need for descriptive and meaningful names						•	•
Complexity and size of the prompts		•					
Authentication to secondary device to access the authentication secret			•				
Continuous hardware connection not required							•
Intermittent Events							
Reauthentication due to user inactivity	•	•	•	•	•	•	•
Fixed periodic reauthentication	•	•	•	•	•	•	•
Provisions for technical assistance	•	•	•	•	•	•	•
Provisions to create and change passwords	•						